

Solution Code - Spring Core and Maven - XML_AOP_Logging

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\main\java\com\library\BookService.java

```
1 | package com.library;
2 |
3 | public class BookService {
4 |     public void getBooks() {
5 |         System.out.println("Executing core business logic: Fetching books catalog.");
6 |     }
7 | }
```

Source File: src\main\java\com\library\LibraryApp.java

```
1 | package com.library;
2 |
3 | import org.springframework.context.ApplicationContext;
4 | import org.springframework.context.support.ClassPathXmlApplicationContext;
5 |
6 | public class LibraryApp {
7 |     public static void main(String[] args) {
8 |         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
9 |         BookService service = (BookService) context.getBean("bookService");
10 |         service.getBooks();
11 |     }
12 | }
```

Source File: src\main\java\com\library\LoggingAspect.java

```
1 | package com.library;
2 |
3 | public class LoggingAspect {
4 |     public void beforeAdvice() {
5 |         System.out.println("[XML-AOP-LOG] Before Advice executed: Logged service invocation.");
6 |     }
7 | }
```

Source File: src\main\resources\applicationContext.xml

```
1 | <beans xmlns="http://www.springframework.org/schema/beans"
2 |       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3 |       xmlns:aop="http://www.springframework.org/schema/aop"
4 |       xsi:schemaLocation="http://www.springframework.org/schema/beans
5 |       http://www.springframework.org/schema/beans/spring-beans.xsd
6 |       http://www.springframework.org/schema/aop
7 |       http://www.springframework.org/schema/aop/spring-aop.xsd">
```

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```
8 |  
9 |     <bean id="bookService" class="com.library.BookService" />  
10 |     <bean id="loggingAspect" class="com.library.LoggingAspect" />  
11 |  
12 |     <aop:config>  
13 |         <aop:aspect ref="loggingAspect">  
14 |             <aop:before method="beforeAdvice" pointcut="execution(* com.library.BookService.get*(..))" />  
15 |         </aop:aspect>  
16 |     </aop:config>  
17 | </beans>
```